MILITARY SPECIFICATION SHEET

CRYSTAL UNITS, QUARTZ, CR-40/U

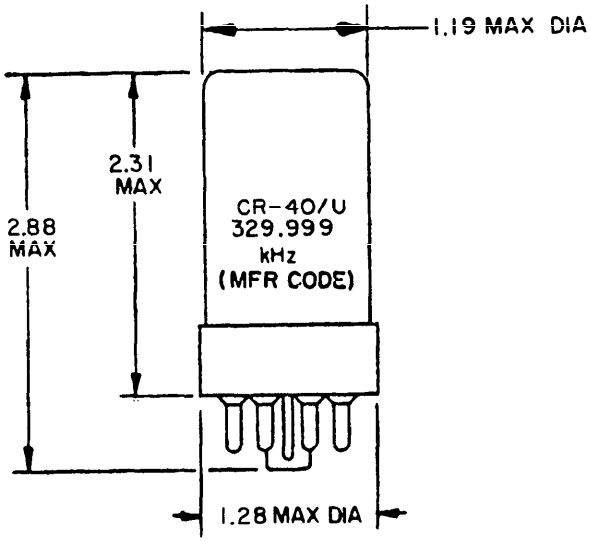
This specification is approved for use by all Departments and Agencies of the Department of Defense.

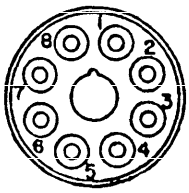
The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of specification MIL-C-3098.

Pertinent characteristics: 160 to 330 kHz; fundamental; controlled; series.

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INCHES	MM
1.19	30.23
1.28	32.51
2.31	58.67
2.88	73.15





NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Base dimensions shall be in accordance with mating-base dimensions of Specification MIL-S-12883/1.
4. Pins 2 and 4 shall be connected internally; pins 6 and 8 shall be connected internally.
5. The quartz resonator shall be connected between pins (2,4) and pins (6,8).

FIGURE 1. Dimensions and configuration.

© denotes changes

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-15/U (See MIL-H-10056/6)
Dimensions, marking, and configuration	See figure.
Frequency range	160 to 330 kHz inclusive
Capacitance, shunt: kHz	pF max
160 to 250	$\frac{4,320}{f} + 2$
250 + to 330	$\frac{1,110}{f} + 2$
Ⓒ Frequency tolerance	±30 PPM
Ⓒ Frequency stability	±5 PPM
Equivalent resistance: 1/ kHz	Ohms, maximum
160 to 250	150
250+ to 330	600
Mode of oscillation	Fundamental
Reference temperature	70° ±1°C
Temperature ranges:	
Operating	65° to 75°C inclusive.
Operable	-55° to +65°C inclusive
Test set, calibration values, and rated drive level	See table II
Shock:	
Ⓒ Frequency change permitted	± 1.5 PPM
Equivalent resistance change permitted	±15 percent
Vibration: 2/	
Ⓒ Frequency change permitted	±1.5 PPM
Equivalent resistance change permitted	±15 percent
Ⓒ Thermal shock:	
Frequency change permitted	±10 PPM
Equivalent resistance change permitted	±15 percent
Bond strength: kHz	Grams, minimum
160 to 250	1,000
250+ to 330	800

1/ Throughout the operating temperature range, the maximum resistance measurement shall be not greater than 150 percent of the minimum resistance measurement.
2/ Method 201 of MIL-STD-202, amplitude 0.015 inch (total excursion 0.030 inch).

TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	kHz	ohms	mA	volt	mW
TS-710/TSM	160 to 250	100	---	0.10	0.1
	250+ to 330	420	---	0.20	0.1

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NOTES:
1. This crystal unit type is to be used for special applications and for replacement purposes only. Specific approval of the Government is required prior to each intended use. See MIL-STD-683.
2. Qualification is not required. First article inspection is applicable.
3. Pin alinement and frequency and resistance at room temperature (for controlled types) requirements are not applicable.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Army -
Navy - AS, CG, MC, OS
Air Force - 17, 19

Preparing activity:
Army - EL

Agent:
DLA - ES

(Project 5955-0512-7)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-C-3098/20C CRYSTAL UNITS, QUARTZ, CR-40/U

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